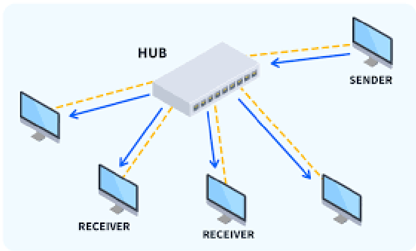
# Computer Network

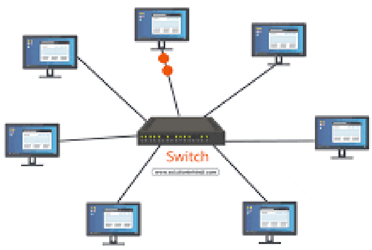
## Practical 1 : Study of different Network devices.

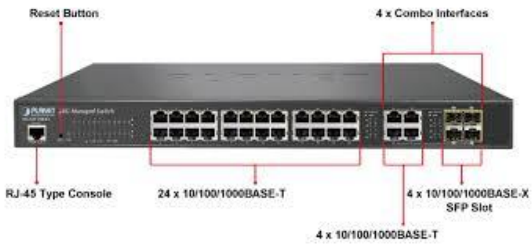
### 1. Hub




A hub is a standard network point, also referred to as a network hub, to connect devices in a network. The hub has numerous ports. If a packet makes it to one port, it can be viewed by all the network segments because a packet is duplicated to another port. A hub is a unique device used to enhance a network by enabling increased workstations.

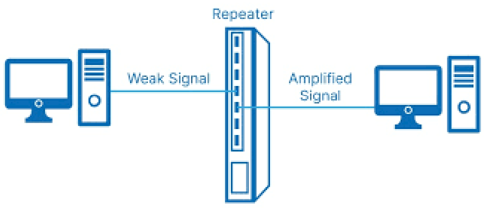
### 2. Switch





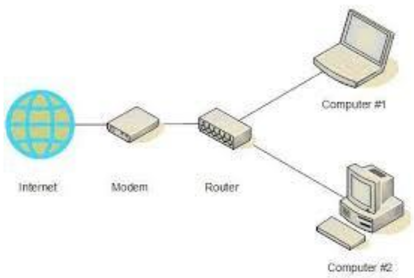

Switches are networking devices operating at layer 2 or a data link layer of the OSI model. They connect devices in a network and use packet switching to send, receive or forward data packets or data frames over the network. A switch has many ports, to which computers are plugged in. When a data frame arrives at any port of a network switch, it examines the destination address, performs necessary checks and sends the frame to the corresponding device(s). It supports unicast, multicast as well as broadcast communications.

## 3. Repeater



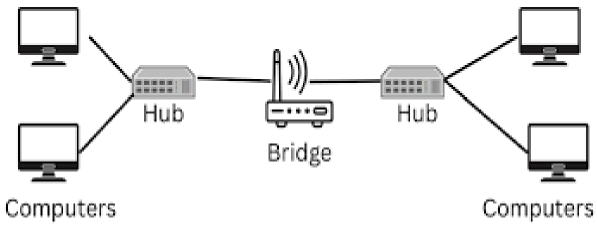

When an electrical signal is transmitted via a channel, it gets attenuated depending upon the nature of the channel or the technology. This poses a limitation upon the length of the LAN or coverage area of cellular networks. This problem is alleviated by installing repeaters at certain intervals. Repeaters amplifies the attenuated signal and then retransmits it. Digital repeaters can even reconstruct signals distorted by transmission loss. So, repeaters are popularly incorporated to connect between two LANs thus forming a large single LAN.

## 4. Router



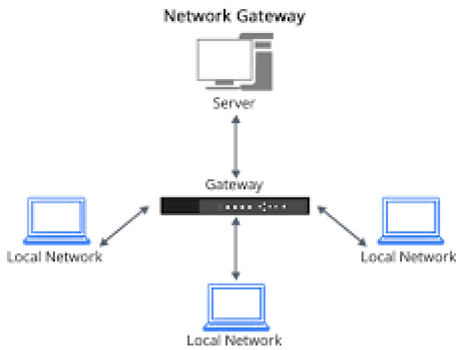

Routers are networking devices operating at layer 3 or a network layer of the OSI model. They are responsible for receiving, analyzing, and forwarding data packets among the connected computer networks. When a data packet arrives, the router inspects the destination address, consults its routing tables to decide the optimal route and then transfers the packet along this route.

## 5. Bridge




A bridge is a networking device that connects two LANs or subnetworks using the same protocol. Unlike a repeater, which forwards and amplifies all packets, a bridge intelligently filters network traffic. It only forwards packets that are intended for devices on the other network segment. This reduces unnecessary traffic and improves network performance. By combining two LANs into an extended LAN, a bridge provides better efficiency and traffic management.

## 6. Gateway




A gateway is a network device that connects two different networks and enables communication between them. It acts as a protocol converter by translating data between networks that use different communication protocols. All data entering or leaving a network passes through the gateway, making it the network's entry and exit point. Gateways monitor and control incoming and outgoing traffic to ensure smooth communication. They are essential for connecting dissimilar networks and facilitating data exchange.